

# Esp32Tap — Depth-First Cluster Build and Test

**Guide mode:** Empty-board build · Build one cluster, prove it, sign it, then unlock the next.

No treadmill cable before clusters 1–10 pass; do not connect a treadmill cable until Cluster 11. No exact physical placement is prescribed. Follow the named parts, pins, nets, and colors.

|                           |                  |
|---------------------------|------------------|
| <b>Device serial:</b>     | <b>Operator:</b> |
| <b>Hardware revision:</b> | <b>Date:</b>     |

**Source manifest:** esp32tap-breadboard-netlist.csv; esp32tap-breadboard-clusters.html; esp32tap-breadboard-from-to.html; DigiKey invoice 129832848; superseded module checklist used only for already-proven limits and sequences.

## Source contract

**Mutually exclusive source rule:** USB and STANDALONE POWER are mutually exclusive—one source only. USB connected means STANDALONE POWER physically removed. STANDALONE POWER installed means USB physically disconnected. All sources off before any RJ45 attachment/removal, harness change, or direct-path change. COIL POWER stays open through the unloaded test.

## Safety contract

- Initial source: 8.00 V, initial 250 mA current limit; coil-open current below 50 mA. VIN must be 7.20–7.90 V and LOGIC\_3V3 must be 3.20–3.40 V.
- TREAD\_OK is low below UV, high at 8.00 V, and low above OV. UV rising is 6.25–6.55 V; OV falling is 10.30–10.90 V.
- TPS709 enabled: 4.75–5.25 V; TPS709 disabled: below 0.25 V. Loaded relay current limit no more than 500 mA.
- Coil: 90–110 mA and at least 4.50 V. BC337 VCE no more than 0.30 V. Feedback (1,0) energized and (0,1) bypass; 00 or 11 is a fault.
- Removal of USB logic power, GPIO21 command, TREAD\_OK, or VIN must restore NC bypass within 100 ms.
- Five-minute hold: TPS709 and BC337 no more than 45°C and no more than 10°C over ambient.
- Any unexpected reset is a STOP. No treadmill relay transfer or MCU TX is permitted by this guide.

**Firmware:** The current no-control diagnostic cannot pass a relay exercise gate and cannot pass a UART exercise gate. Relay/coil/TX work requires a bounded relay exerciser with exact identity. Standalone/bypass requires an observer identity and manifest proving relay and TX disabled.

Wire palette: BLACK GND · RED +8V\_RAW/FUSED\_8V · ORANGE VIN · BLUE TSR\_3V3/LOGIC\_3V3 · VIOLET +5V\_RLY/coil · YELLOW control/UART · GREEN feedback/RX · WHITE direct pass-through.

## Cluster 1 — Raw protection

**Dependencies:** Empty board and verified bench equipment  
**Inputs:** +8V\_RAW, GND

**Outputs:** VIN, FUSED\_8V  
**Source state:** USB absent; STANDALONE POWER removed;  
COIL POWER removed; bench 8.00 V, initial 250 mA limit

### Parts

**F1** RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar. **D1** 1N5822-TP Schottky · 1N5822-TPMSCT-ND · cathode stripe faces VIN. **D2** P6KE12A-TP · P6KE12A-TPMSCT-ND · cathode to VIN. **CIN** 22  $\mu$ F 25 V · 732-8821-1-ND · negative stripe to GND.

**Operator part record:** F1 · RXEF075 · 750 mA hold · RXEF075HF-ND · non-polar

**Operator wiring record:** F1 · lead 1 · +8V\_RAW · RED; F1 · lead 2 · FUSED\_8V · RED; D1 · anode · FUSED\_8V · RED; D1 · cathode · VIN · ORANGE; D2 · cathode · VIN · ORANGE; D2 · anode · GND · BLACK; CIN · positive · VIN · ORANGE; CIN · negative · GND · BLACK

### Build — exact pin/net/color wiring

| Step | Ref | Color  | From                | To                  | Part | Note                                                                                                      |
|------|-----|--------|---------------------|---------------------|------|-----------------------------------------------------------------------------------------------------------|
| 1    | 15  | RED    | +8V_RAW             | RXEF075 lead 1      | Wire | Local electronics tap only<br>Connect F1 lead 1 to +8V_RAW using RED.                                     |
| 2    | 16  | RED    | RXEF075 lead 2      | 1N5822 anode        | Wire | Non-striped diode end<br>Connect F1 lead 2 to FUSED_8V using RED. Connect D1 anode to FUSED_8V using RED. |
| 3    | 17  | ORANGE | 1N5822 cathode      | VIN                 | Wire | Striped diode end<br>Connect D1 cathode to VIN using ORANGE.                                              |
| 4    | 18  | ORANGE | VIN                 | P6KE12A cathode     | Wire | Striped TVS end<br>Connect D2 cathode to VIN using ORANGE.                                                |
| 5    | 19  | BLACK  | P6KE12A anode       | GND                 | Wire | TVS return<br>Connect D2 anode to GND using BLACK.                                                        |
| 6    | 23  | ORANGE | VIN                 | 22 $\mu$ F positive | Wire | Bulk capacitor<br>Connect CIN positive to VIN using ORANGE.                                               |
| 7    | 24  | BLACK  | 22 $\mu$ F negative | GND                 | Wire | Observe polarity<br>Connect CIN negative to GND using BLACK.                                              |

### Unpowered test

| Instruction / kind / points / expected                                                                                              | Evidence |
|-------------------------------------------------------------------------------------------------------------------------------------|----------|
| Check polarity and resistance · polarity · D1 cathode/VIN, D2 cathode/VIN, CIN negative/GND, VIN–GND · stripes correct and no short |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                         | Measured |
|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Apply protected raw power · +8V_RAW at 8.00 V · FUSED_8V and VIN · record volts/current · VIN 7.20–7.90 V; coil-open current below 50 mA |          |

**STOP gate:** Current approaches 250 mA, VIN is outside 7.20–7.90 V, polarity is wrong, or anything warms. Remove power; do not build Cluster 2.

**PASS gate:** Protection chain, polarity, VIN window, and coil-open current pass. **PASS — continue** to Cluster 2 only.

Operator:

Date:

Signed PASS:

## Cluster 2 — TSR supply

**Dependencies:** Cluster 1  
**Inputs:** VIN, GND

**Outputs:** TSR\_3V3  
**Source state:** Cluster 1 source state retained; STANDALONE POWER remains removed

### Parts

**U2** TSR 1-2433E · 3.3 V converter · 1951-TSR1-2433E-ND · pin order verified. **C2/C3** 10  $\mu$ F 50 V electrolytics · 1189-2322-ND · negative stripes to GND. **JP1** removable STANDALONE POWER · NOT ORDERED · leave removed.

**Operator part record:** U2/C2/C3/JP1 · TSR 1-2433E; two electrolytics; STANDALONE POWER · 3.3 V; 10  $\mu$ F 50 V; removable link · 1951-TSR1-2433E-ND; 1189-2322-ND; NOT ORDERED · U2 pin order verified; capacitor negatives to GND; link removed

**Operator wiring record:** U2 · pin 1 · VIN · ORANGE; C2 · positive · VIN · ORANGE; U2 · pin 2 · GND · BLACK; C2 · negative · GND · BLACK; U2 · pin 3 · TSR\_3V3 · BLUE; C3 · positive · TSR\_3V3 · BLUE; C3 · negative · GND · BLACK; JP1 · between · TSR\_3V3 and LOGIC\_3V3 · BLUE

### Build — exact pin/net/color wiring

| Step | Ref | Color  | From                           | To                             | Part                       | Note                                                                                                       |
|------|-----|--------|--------------------------------|--------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------|
| 1    | 20  | ORANGE | VIN                            | TSR1-2433E pin 1               | Wire                       | Converter input<br>Connect U2 pin 1 to VIN using ORANGE.                                                   |
| 2    | 25  | ORANGE | VIN                            | 10 $\mu$ F input-cap positive  | Wire                       | Near TSR input<br>Connect C2 positive to VIN using ORANGE.                                                 |
| 3    | 21  | BLACK  | TSR1-2433E pin 2               | GND                            | Wire                       | Converter ground<br>Connect U2 pin 2 to GND using BLACK.                                                   |
| 4    | 26  | BLACK  | 10 $\mu$ F input-cap negative  | GND                            | Wire                       | Observe polarity<br>Connect C2 negative to GND using BLACK.                                                |
| 5    | 22  | BLUE   | TSR1-2433E pin 3               | TSR_3V3                        | Wire                       | Converter output<br>Connect U2 pin 3 to TSR_3V3 using BLUE.                                                |
| 6    | 27  | BLUE   | TSR_3V3                        | 10 $\mu$ F output-cap positive | Wire                       | Near TSR output<br>Connect C3 positive to TSR_3V3 using BLUE.                                              |
| 7    | 28  | BLACK  | 10 $\mu$ F output-cap negative | GND                            | Wire                       | Observe polarity<br>Connect C3 negative to GND using BLACK.                                                |
| 8    | 31  | BLUE   | TSR_3V3                        | LOGIC_3V3                      | STANDALONE<br>POWER jumper | Leave removed when USB is attached<br>Fit JP1 between TSR_3V3 and LOGIC_3V3 using BLUE; leave JP1 removed. |

### Unpowered test

| Instruction / kind / points / expected                                                                                                              | Evidence |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Verify converter map and capacitor polarity · continuity · U2 pins 1/2/3 to VIN/GND/TSR_3V3 · exact mapping; TSR_3V3 not joined to future LOGIC_3V3 |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                         | Measured |
|--------------------------------------------------------------------------------------------------------------------------|----------|
| Energize verified VIN · VIN · TSR_3V3 · record input/output/current · TSR_3V3 3.20–3.40 V and future LOGIC_3V3 unpowered |          |

**STOP gate:** TSR\_3V3 is outside 3.20–3.40 V, polarity is wrong, LOGIC\_3V3 is energized across the removed link, or current/temperature rises.

**PASS gate:** Verified Cluster 1 VIN produces isolated TSR\_3V3 in range. **PASS — continue** to Cluster 3 only.

Operator:

Date:

Signed PASS:

## Cluster 3 — DevKit and logic supply

**Dependencies:** Cluster 2; Verified USB bench source  
**Inputs:** TSR\_3V3, USB\_5V, GND

**Outputs:** LOGIC\_3V3, DevKit GND, GPIO21\_CMD\_POST, GPIO15\_TX\_ENABLE\_POST, GPIO17\_TX, GPIO38, USB\_5V  
**Source state:** First USB-only with STANDALONE POWER removed; then USB physically disconnected before standalone link installation

### Parts

**U1** ESP32-S3-DEVKITC-1-N8R8 · 1965-ESP32-S3-DEVKITC-1-N8R8-ND · antenna clear, USB ports identified. **JP1** STANDALONE POWER · NOT ORDERED · removable and labeled. **C4** 100 nF · BC1084CT-ND · non-polar.

**Operator part record:** U1/JP1/C4 · ESP32-S3-DEVKITC-1-N8R8; STANDALONE POWER; capacitor · DevKit; removable link; 100 nF · 1965-ESP32-S3-DEVKITC-1-N8R8-ND; NOT ORDERED; BC1084CT-ND · antenna clear; USB ports identified; link removable; capacitor non-polar

**Operator wiring record:** DevKit · 3V3 · LOGIC\_3V3 · BLUE; DevKit · GND · GND · BLACK; C4 · lead 1 · LOGIC\_3V3 · BLUE; C4 · lead 2 · GND · BLACK; TSR\_3V3 · endpoint · the JP1 standalone-side post · BLUE; LOGIC\_3V3 · endpoint · the JP1 logic-side post · BLUE; DevKit · GPIO15 · the DevKit-side GPIO15 jumper post · YELLOW; DevKit · GPIO21 · the DevKit-side GPIO21 jumper post · YELLOW

### Build — exact pin/net/color wiring

| Step | Ref      | Color  | From               | To                                 | Part        | Note                                                                                      |
|------|----------|--------|--------------------|------------------------------------|-------------|-------------------------------------------------------------------------------------------|
| 1    | 29       | BLUE   | DevKit 3V3         | LOGIC_3V3                          | Jumper wire | Logic rail<br>Connect DevKit 3V3 to LOGIC_3V3 using BLUE.                                 |
| 2    | 30       | BLACK  | DevKit GND         | GND                                | Jumper wire | Common ground<br>Connect DevKit GND to GND using BLACK.                                   |
| 3    | NEW C3-3 | BLUE   | C4 lead 1          | LOGIC_3V3                          | Wire        | Connect C4 lead 1 to LOGIC_3V3 using BLUE.                                                |
| 4    | NEW C3-4 | BLACK  | C4 lead 2          | GND                                | Wire        | Connect C4 lead 2 to GND using BLACK.                                                     |
| 5    | NEW C3-5 | BLUE   | TSR_3V3 endpoint   | the JP1 standalone-side post       | Wire        | Connect TSR_3V3 endpoint to the JP1 standalone-side post using BLUE.                      |
| 6    | NEW C3-6 | BLUE   | LOGIC_3V3 endpoint | the JP1 logic-side post            | Wire        | Connect LOGIC_3V3 endpoint to the JP1 logic-side post using BLUE; leave JP1 removed.      |
| 7    | NEW C3-7 | YELLOW | DevKit GPIO15      | the DevKit-side GPIO15 jumper post | Wire        | Connect DevKit GPIO15 to the DevKit-side GPIO15 jumper post using YELLOW; label the post. |
| 8    | NEW C3-8 | YELLOW | DevKit GPIO21      | the DevKit-side GPIO21 jumper post | Wire        | Connect DevKit GPIO21 to the DevKit-side GPIO21 jumper post using YELLOW; label the post. |

### Unpowered test

| Instruction / kind / points / expected                                                                                  | Evidence |
|-------------------------------------------------------------------------------------------------------------------------|----------|
| Map source handoff · mapping · DevKit 3V3, LOGIC_3V3, TSR_3V3, JP1 · USB path mapped; JP1 removable; no source backfeed |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                              | Measured |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Prove each exclusive source · USB_5V then TSR_3V3 · LOGIC_3V3 · record both runs · LOGIC_3V3 3.20–3.40 V; never energize USB and JP1 together |          |

**STOP gate:** Sources overlap, either source backfeeds the other, LOGIC\_3V3 is outside 3.20–3.40 V, or DevKit boot is unstable.

**PASS gate:** USB-only and standalone-only handoff both pass, with the other source physically absent. **PASS — continue** to Cluster 4 only.

Operator:

Date:

Signed PASS:

## Cluster 4 — TPS3700 voltage monitor

**Dependencies:** Cluster 1, Cluster 3  
**Inputs:** VIN, LOGIC\_3V3, GND

**Outputs:** UV\_SENSE, OV\_SENSE, TREAD\_OK, TREAD\_OK\_MCU  
**Source state:** USB logic only; STANDALONE POWER removed; bench source swept at protected VIN

### Parts

**U3** TPS3700DDCR · 296-30395-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **RUV** 150 kΩ · 150KXBK-ND; **ROV** 255 kΩ · 13-MFR-25FBF52-255K-ND; 10 kΩ/100 kΩ/4.7 kΩ · NOT ORDERED; two 1 nF · 445-173473-1-ND; 100 nF · BC1084CT-ND.

**Adapter gate:** continuity-map adapter pins 1–6 to the header before installation. Record the mapped header positions: 1 \_\_\_\_ 2 \_\_\_\_ 3 \_\_\_\_ 4 \_\_\_\_ 5 \_\_\_\_ 6 \_\_\_\_.

**Divider formulas:**  $UV\_SENSE = VIN \times 10\text{ k}\Omega / (150\text{ k}\Omega + 10\text{ k}\Omega) = VIN / 16$ .  $OV\_SENSE = VIN \times 10\text{ k}\Omega / (255\text{ k}\Omega + 10\text{ k}\Omega) = VIN / 26.5$ .

**Operator part record:** U3/dividers/filters · TPS3700DDCR plus adapter and UV/OV networks · 150 kΩ/10 kΩ; 255 kΩ/10 kΩ; 1 nF; 100 nF · 296-30395-1-ND; A879AR-ND; 150KXBK-ND; 13-MFR-25FBF52-255K-ND; 445-173473-1-ND; BC1084CT-ND; NOT ORDERED · meter-map adapter pin 1; capacitors non-polar

**Operator wiring record:** See numbered point-to-point steps below.

**Operator header record:** HDR3 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

### Build — exact pin/net/color wiring

| Step | Ref | Color  | From                   | To                             | Part | Note                                                                                                                                                                  |
|------|-----|--------|------------------------|--------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 32  | ORANGE | VIN                    | TPS3700 pin 5 VDD              | Wire | Supply<br>Connect U3 pin 5 to VIN using ORANGE; record initials ____.                                                                                                 |
| 2    | 33  | BLACK  | TPS3700 pin 2          | GND                            | Wire | Ground<br>Connect U3 pin 2 to GND using BLACK; record initials ____.                                                                                                  |
| 3    | 36  | WHITE  | UV_SENSE               | TPS3700 pin 3 INA+             | Wire | UV input<br>Connect U3 pin 3 to UV_SENSE using WHITE; record initials ____.                                                                                           |
| 4    | 43  | WHITE  | OV_SENSE               | TPS3700 pin 4 INB-             | Wire | OV input<br>Connect U3 pin 4 to OV_SENSE using WHITE; record initials ____.                                                                                           |
| 5    | 48  | YELLOW | TPS3700 pin 1 OUTA     | TREAD_OK                       | Wire | Open-drain output<br>Connect U3 pin 1 to TREAD_OK using YELLOW; record initials ____.                                                                                 |
| 6    | 49  | YELLOW | TPS3700 pin 6 OUTB     | TREAD_OK                       | Wire | Tie outputs together<br>Connect U3 pin 6 to TREAD_OK using YELLOW; record initials ____.                                                                              |
| 7    | 34  | ORANGE | VIN                    | R_UV_TOP 150 kΩ, end 1         | Wire | UV divider<br>Connect R_UV_TOP 1 to VIN using ORANGE; record initials ____.                                                                                           |
| 8    | 35  | WHITE  | R_UV_TOP end 2         | UV_SENSE                       | Wire | Divider midpoint<br>Connect R_UV_TOP 2 to UV_SENSE using WHITE; record initials ____.                                                                                 |
| 9    | 37  | WHITE  | UV_SENSE               | R_UV_BOTTOM 10 kΩ, end 1       | Wire | Divider bottom<br>Connect R_UV_BOTTOM 1 to UV_SENSE using WHITE; record initials ____.                                                                                |
| 10   | 38  | BLACK  | R_UV_BOTTOM end 2      | GND                            | Wire | Divider return<br>Connect R_UV_BOTTOM 2 to GND using BLACK; record initials ____.                                                                                     |
| 11   | 39  | WHITE  | UV_SENSE               | C_UV 1 nF, end 1               | Wire | Filter<br>Connect C_UV 1 to UV_SENSE using WHITE; record initials ____.                                                                                               |
| 12   | 40  | BLACK  | C_UV end 2             | GND                            | Wire | Filter return<br>Connect C_UV 2 to GND using BLACK; record initials ____.                                                                                             |
| 13   | 41  | ORANGE | VIN                    | R_OV_TOP 255 kΩ, end 1         | Wire | OV divider<br>Connect R_OV_TOP 1 to VIN using ORANGE; record initials ____.                                                                                           |
| 14   | 42  | WHITE  | R_OV_TOP end 2         | OV_SENSE                       | Wire | Divider midpoint<br>Connect R_OV_TOP 2 to OV_SENSE using WHITE; record initials ____.                                                                                 |
| 15   | 44  | WHITE  | OV_SENSE               | R_OV_BOTTOM 10 kΩ, end 1       | Wire | Divider bottom<br>Connect R_OV_BOTTOM 1 to OV_SENSE using WHITE; record initials ____.                                                                                |
| 16   | 45  | BLACK  | R_OV_BOTTOM end 2      | GND                            | Wire | Divider return<br>Connect R_OV_BOTTOM 2 to GND using BLACK; record initials ____.                                                                                     |
| 17   | 46  | WHITE  | OV_SENSE               | C_OV 1 nF, end 1               | Wire | Filter<br>Connect C_OV 1 to OV_SENSE using WHITE; record initials ____.                                                                                               |
| 18   | 47  | BLACK  | C_OV end 2             | GND                            | Wire | Filter return<br>Connect C_OV 2 to GND using BLACK; record initials ____.                                                                                             |
| 19   | 50  | BLUE   | LOGIC_3V3              | R_TREAD_PULLUP 10 kΩ, end 1    | Wire | Pull-up<br>Connect R_TREAD_PULLUP 1 to LOGIC_3V3 using BLUE; record initials ____.                                                                                    |
| 20   | 51  | YELLOW | R_TREAD_PULLUP end 2   | TREAD_OK                       | Wire | Pull-up node<br>Connect R_TREAD_PULLUP 2 to TREAD_OK using YELLOW; record initials ____.                                                                              |
| 21   | 52  | YELLOW | TREAD_OK               | R_TREAD_PULLDOWN 100 kΩ, end 1 | Wire | Power-off pull-down<br>Connect R_TREAD_PULLDOWN 1 to TREAD_OK using YELLOW; record initials ____.                                                                     |
| 22   | 53  | BLACK  | R_TREAD_PULLDOWN end 2 | GND                            | Wire | Pull-down return<br>Connect R_TREAD_PULLDOWN 2 to GND using BLACK; record initials ____.                                                                              |
| 23   | 54  | YELLOW | TREAD_OK               | R_TREAD_MCU 4.7 kΩ, end 1      | Wire | GPIO isolation<br>Connect R_TREAD_MCU 1 to TREAD_OK using YELLOW; record initials ____.                                                                               |
| 24   | 55  | YELLOW | R_TREAD_MCU end 2      | DevKit GPIO6                   | Wire | TREAD_OK_MCU<br>Connect R_TREAD_MCU 2 to TREAD_OK_MCU using YELLOW; record initials ____.<br>Connect DEVKIT GPIO6 to TREAD_OK_MCU using YELLOW; record initials ____. |
| 25   | 56  | ORANGE | VIN                    | C_TPS3700 100 nF, end 1        | Wire | Decoupling<br>Connect C_TPS3700 1 to VIN using ORANGE; record initials ____.                                                                                          |
| 26   | 57  | BLACK  | C_TPS3700 end 2        | GND                            | Wire | Decoupling return<br>Connect C_TPS3700 2 to GND using BLACK; record initials ____.                                                                                    |

### Unpowered test

| Instruction / kind / points / expected                                                                                                       | Evidence |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Meter divider and adapter · resistance · VIN–UV_SENSE–GND, VIN–OV_SENSE–GND, U3 pins · 150 kΩ/10 kΩ and 255 kΩ/10 kΩ networks; pin map exact |          |
| Continuity-map adapter header · continuity · adapter pins 1–6 to header · one-to-one recorded mapping before installation                    |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                 | Measured |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Sweep protected VIN · VIN · TREAD_OK · record low/high transitions · low below UV, high at 8.00 V, low above OV; UV rising 6.25–6.55 V; OV falling 10.30–10.90 V |          |

**STOP gate:** TREAD\_OK three-state behavior or thresholds fail; never exceed protected VIN about 11.1 V; stop if TVS draws appreciable current or warms.

**PASS gate:** Both sweep directions and GPIO6 isolated observation match limits. **PASS — continue** to Cluster 5 only.

Operator: \_\_\_\_\_ Date: \_\_\_\_\_ Signed PASS: \_\_\_\_\_  
Cluster 4 · TPS3700 monitor

## Cluster 5 — AHC08 permission logic

**Dependencies:** Cluster 3, Cluster 4  
**Inputs:** LOGIC\_3V3, TREAD\_OK, RELAY\_CMD, TX\_ENABLE, GND

**Outputs:** RELAY\_GATE, TX\_GATE  
**Source state:** USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

**Source state record:** USB logic and bench VIN; remove the corresponding DevKit command jumper before manual injection

### Parts

U4 SN74AHC08N PDIP-14 · 296-4524-5-ND · notch verified, across center channel. **C5** 100 nF · BC1084CT-ND · non-polar. **R1/R2** 10 kΩ command pull-downs · NOT ORDERED.

**Operator part record:** U4/C5/R1/R2 · SN74AHC08N; capacitor; pull-downs · PDIP-14; 100 nF; two 10 kΩ · 296-4524-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar

**Operator wiring record:** See numbered point-to-point steps below.

### Build — exact pin/net/color wiring

| Step | Ref | Color   | From                 | To                          | Part    | Note                                                                                                                                                                                                                                                                 |
|------|-----|---------|----------------------|-----------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 58  | BLUE    | LOGIC_3V3            | AHC08 pin 14                | Wire    | VCC<br>Connect U4 pin 14 to LOGIC_3V3 using BLUE; record initials ____.                                                                                                                                                                                              |
| 2    | 59  | BLACK   | AHC08 pin 7          | GND                         | Wire    | Ground<br>Connect U4 pin 7 to GND using BLACK; record initials ____.                                                                                                                                                                                                 |
| 3    | 60  | YELLOW  | DevKit GPIO21        | AHC08 pin 1                 | Wire    | RELAY_CMD<br>Connect GPIO21 jumper post endpoint to U4 pin 1 / RELAY_CMD using YELLOW; record initials ____.                                                                                                                                                         |
| 4    | 63  | YELLOW  | TREAD_OK             | AHC08 pin 2                 | Wire    | Relay permission<br>Connect U4 pin 2 to TREAD_OK using YELLOW; record initials ____.                                                                                                                                                                                 |
| 5    | 64  | YELLOW  | AHC08 pin 3          | RELAY_GATE                  | Wire    | Relay gate output<br>Connect U4 pin 3 to RELAY_GATE using YELLOW; record initials ____.                                                                                                                                                                              |
| 6    | 65  | YELLOW  | DevKit GPIO15        | AHC08 pin 4                 | Wire    | TX_ENABLE<br>Connect GPIO15 jumper post endpoint to U4 pin 4 / TX_ENABLE using YELLOW; record initials ____.                                                                                                                                                         |
| 7    | 68  | YELLOW  | TREAD_OK             | AHC08 pin 5                 | Wire    | TX permission<br>Connect U4 pin 5 to TREAD_OK using YELLOW; record initials ____.                                                                                                                                                                                    |
| 8    | 69  | YELLOW  | AHC08 pin 6          | TX_GATE                     | Wire    | TX gate output<br>Connect U4 pin 6 to TX_GATE using YELLOW; record initials ____.                                                                                                                                                                                    |
| 9    | 70a | BLACK   | AHC08 pin 9          | GND                         | Wire    | Tie unused inputs low legacy Ref 70; 4 wires<br>Connect U4 9 to GND using BLACK; record initials ____.                                                                                                                                                               |
| 10   | 70b | BLACK   | AHC08 pin 10         | GND                         | Wire    | Tie unused inputs low legacy Ref 70; 4 wires<br>Connect U4 10 to GND using BLACK; record initials ____.                                                                                                                                                              |
| 11   | 70c | BLACK   | AHC08 pin 12         | GND                         | Wire    | Tie unused inputs low legacy Ref 70; 4 wires<br>Connect U4 12 to GND using BLACK; record initials ____.                                                                                                                                                              |
| 12   | 70d | BLACK   | AHC08 pin 13         | GND                         | Wire    | Tie unused inputs low legacy Ref 70; 4 wires<br>Connect U4 13 to GND using BLACK; record initials ____.                                                                                                                                                              |
| 13   | 71a | NO WIRE | AHC08 pin 8          | NO CONNECTION               | NO WIRE | Unused outputs legacy Ref 71; 2 intentional opens<br>Leave U4 pin 8 at NC using NO WIRE; record initials ____.                                                                                                                                                       |
| 14   | 71b | NO WIRE | AHC08 pin 11         | NO CONNECTION               | NO WIRE | Unused outputs legacy Ref 71; 2 intentional opens<br>Leave U4 pin 11 at NC using NO WIRE; record initials ____.                                                                                                                                                      |
| 15   | 61  | YELLOW  | DevKit GPIO21        | R_RELAY_CMD_PD 10 kΩ, end 1 | Wire    | Command pull-down DevKit GPIO21 ↔ removable GPIO21 jumper ↔ RELAY_CMD / AHC08 pin 1.<br>Connect GPIO21 jumper post endpoint to U4 pin 1 / RELAY_CMD using YELLOW; record initials ____.<br>Connect R_RELAY_CMD_PD 1 to RELAY_CMD using YELLOW; record initials ____. |
| 16   | 62  | BLACK   | R_RELAY_CMD_PD end 2 | GND                         | Wire    | Pull-down return<br>Connect R_RELAY_CMD_PD 2 to GND using BLACK; record initials ____.                                                                                                                                                                               |
| 17   | 66  | YELLOW  | DevKit GPIO15        | R_TX_ENABLE_PD 10 kΩ, end 1 | Wire    | Enable pull-down DevKit GPIO15 ↔ removable GPIO15 jumper ↔ TX_ENABLE / AHC08 pin 4.<br>Connect GPIO15 jumper post endpoint to U4 pin 4 / TX_ENABLE using YELLOW; record initials ____.<br>Connect R_TX_ENABLE_PD 1 to TX_ENABLE using YELLOW; record initials ____.  |
| 18   | 67  | BLACK   | R_TX_ENABLE_PD end 2 | GND                         | Wire    | Pull-down return<br>Connect R_TX_ENABLE_PD 2 to GND using BLACK; record initials ____.                                                                                                                                                                               |
| 19   | 72  | BLUE    | AHC08 pin 14         | C_AHC08 100 nF, end 1       | Wire    | Place close to IC<br>Connect U4 pin 14 to LOGIC_3V3 using BLUE; record initials ____.<br>Connect C_AHC08 1 to LOGIC_3V3 using BLUE; record initials ____.                                                                                                            |
| 20   | 73  | BLACK   | C_AHC08 end 2        | AHC08 pin 7                 | Wire    | Decoupling return<br>Connect U4 pin 7 to GND using BLACK; record initials ____.<br>Connect C_AHC08 2 to GND using BLACK; record initials ____.                                                                                                                       |

### Unpowered test

| Instruction / kind / points / expected                                                                                                     | Evidence |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Check DIP orientation and control mapping · mapping · U4 pins 1–14 and named nets · exact map, unused inputs grounded, outputs not shorted |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                   | Measured |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Exercise both AND gates · RELAY_CMD/TX_ENABLE plus TREAD_OK · RELAY_GATE/TX_GATE · record 00,10,01,11 truth tables · output high only for command=1 and TREAD_OK=1 |          |

Remove GPIO15 jumper before manual injection of TX\_ENABLE. Remove GPIO21 jumper before manual injection of RELAY\_CMD. Never drive against a connected DevKit GPIO.

**STOP gate:** Any gate rises without both inputs, a command GPIO remains connected during manual drive, or supply/reset behavior is abnormal.

**PASS gate:** Both four-state truth tables pass and temporary drives are removed. **PASS — continue** to Cluster 6 only.

Operator:

Date:

Signed PASS:

## Cluster 6 — TPS709 and BC337 driver

**Dependencies:** Cluster 1, Cluster 5  
**Inputs:** VIN, RELAY\_GATE, GND

**Outputs:** +5V\_RLY, RELAY\_COIL–  
**Source state:** Valid VIN/TREAD\_OK; COIL POWER remains removed for the unloaded test

### Parts

**U5** TPS70950DBVR · 296-35484-1-ND; **adapter** LCQT-SOT23-6 · A879AR-ND; **header** cut from PRPC040SAAN-RC · S1011EC-40-ND; **Q1** BC337-40 · 4878-BC337-40CT-ND; **RBASE** 560  $\Omega$  · MFR-25FBF52-560R-ND; 10 k $\Omega$  · NOT ORDERED; 1  $\mu$ F · 445-173583-1-ND; 4.7  $\mu$ F · 732-8660-1-ND; **JP2** COIL POWER · NOT ORDERED.

**Operator part record:** U5/Q1/RBASE/caps/JP2 · TPS70950DBVR; BC337-40; 560  $\Omega$ /10 k $\Omega$ ; 1  $\mu$ F/4.7  $\mu$ F; COIL POWER · 5 V regulator and unloaded driver · 296-35484-1-ND; A879AR-ND; 4878-BC337-40CT-ND; MFR-25FBF52-560R-ND; 445-173583-1-ND; 732-8660-1-ND; NOT ORDERED · adapter and Q1 leads mapped; electrolytic negative to GND; link removed

**Operator wiring record:** See numbered point-to-point steps below.

**Operator header record:** HDR5 · PRPC040SAAN-RC header · cut header · S1011EC-40-ND · map before installation

### Build — exact pin/net/color wiring

| Step | Ref       | Color   | From                  | To                                | Part | Note                                                                                             |
|------|-----------|---------|-----------------------|-----------------------------------|------|--------------------------------------------------------------------------------------------------|
| 1    | 74        | ORANGE  | VIN                   | TPS709 pin 1 IN                   | Wire | Regulator input<br>Connect U5 pin 1 to VIN using ORANGE; record initials ____.                   |
| 2    | 75        | BLACK   | TPS709 pin 2          | GND                               | Wire | Ground<br>Connect U5 pin 2 to GND using BLACK; record initials ____.                             |
| 3    | 76        | YELLOW  | RELAY_GATE            | TPS709 pin 3 EN                   | Wire | Hardware-gated enable<br>Connect U5 pin 3 to RELAY_GATE using YELLOW; record initials ____.      |
| 4    | 78        | VIOLET  | TPS709 pin 5 OUT      | +5V_RLY                           | Wire | Relay supply<br>Connect U5 pin 5 to +5V_RLY using VIOLET; record initials ____.                  |
| 5    | 77        | NO WIRE | TPS709 pin 4          | NO CONNECTION                     | —    | NC pin<br>Leave U5 pin 4 at NC using NO WIRE; record initials ____.                              |
| 6    | 79        | ORANGE  | VIN                   | C_TPS709_IN 1 $\mu$ F, end 1      | Wire | Input bypass<br>Connect C_TPS709_IN 1 to VIN using ORANGE; record initials ____.                 |
| 7    | 80        | BLACK   | C_TPS709_IN end 2     | GND                               | Wire | Bypass return<br>Connect C_TPS709_IN 2 to GND using BLACK; record initials ____.                 |
| 8    | 81        | VIOLET  | +5V_RLY               | C_TPS709_OUT 4.7 $\mu$ F positive | Wire | Output capacitor<br>Connect C_TPS709_OUT positive to +5V_RLY using VIOLET; record initials ____. |
| 9    | 82        | BLACK   | C_TPS709_OUT negative | GND                               | Wire | Output return<br>Connect C_TPS709_OUT negative to GND using BLACK; record initials ____.         |
| 10   | 83        | YELLOW  | RELAY_GATE            | R_BASE 560 $\Omega$ , end 1       | Wire | Base resistor<br>Connect R_BASE 1 to RELAY_GATE using YELLOW; record initials ____.              |
| 11   | 84        | YELLOW  | R_BASE end 2          | BC337 base                        | Wire | Verify physical lead<br>Connect R_BASE 2 to Q_BASE using YELLOW; record initials ____.           |
| 12   | 85        | YELLOW  | BC337 base            | R_BASE_PD 10 k $\Omega$ , end 1   | Wire | Base pull-down<br>Connect Q1 base to Q_BASE using YELLOW; record initials ____.                  |
| 13   | 86        | BLACK   | R_BASE_PD end 2       | GND                               | Wire | Pull-down return<br>Connect R_BASE_PD 1 to Q_BASE using YELLOW; record initials ____.            |
| 14   | 87        | BLACK   | BC337 emitter         | GND                               | Wire | Connect Q1 emitter to GND using BLACK; record initials ____.                                     |
| 15   | NEW C6-16 | VIOLET  | Q1 collector          | RELAY_COIL–                       | Wire | Verify physical lead<br>Connect Q1 collector to RELAY_COIL– using VIOLET; record initials ____.  |
| 16   | NEW C6-17 | VIOLET  | JP2 supply side       | +5V_RLY                           | Wire | Connect Q1 collector to RELAY_COIL– using VIOLET; record initials ____.                          |
| 17   | NEW C6-18 | VIOLET  | JP2 coil side         | RELAY_COIL+                       | Wire | Connect JP2 supply side to +5V_RLY using VIOLET; record initials ____.                           |

### Unpowered test

| Instruction / kind / points / expected                                                                                             | Evidence |
|------------------------------------------------------------------------------------------------------------------------------------|----------|
| Verify adapter, transistor leads, and isolation · polarity · U5 pins, Q1 B/C/E, output capacitor, JP2 · exact map; COIL POWER open |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                             | Measured |
|------------------------------------------------------------------------------------------------------------------------------|----------|
| Test gate unloaded · RELAY_GATE · +5V_RLY/Q1 base · record on/off · TPS709 enabled 4.75–5.25 V; TPS709 disabled below 0.25 V |          |

**STOP gate:** COIL POWER is installed, enabled or disabled limits fail, output does not discharge below 0.25 V, or Q1/U5 heats.

**PASS gate:** Unloaded regulator and base drive pass with coil isolated. **PASS — continue** to Cluster 7 only.

Operator:

Date:

Signed PASS:



## Cluster 7 — Relay coil, local contacts, and feedback

**Dependencies:** Cluster 6, Cluster 3  
**Inputs:** +5V\_RLY, RELAY\_COIL-, LOGIC\_3V3, GND

**Outputs:** LOCAL\_NC, LOCAL\_NO, FB\_NC, FB\_NO  
**Source state:** Local-only bench exercise; treadmill/RJ45 cables absent; bounded relay exerciser verified; loaded relay limit no more than 500 mA

### Parts

**K1** G5V-2 DC5 DPDT · 39-G5V-2DC5-ND · bottom-view drawing, meter-map physical pins. **D3** P6KE6.8CA bidirectional · 18-P6KE6.8CACT-ND. **JP2** COIL POWER · NOT ORDERED. **RFB** two 10 k $\Omega$  · NOT ORDERED.

**Operator part record:** K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 k $\Omega$  · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

**Operator wiring record:** See numbered point-to-point steps below.

### Build — exact pin/net/color wiring

| Step | Ref      | Color  | From             | To                            | Part              | Note                                                                                                                                                                 |
|------|----------|--------|------------------|-------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 89       | VIOLET | +5V_RLY          | K1 pin 1                      | COIL POWER jumper | Leave removed initially<br>Connect K1 pin 1 to RELAY_COIL+ using VIOLET; record initials ____.                                                                       |
| 2    | 88       | VIOLET | BC337 collector  | K1 pin 16                     | Wire              | RELAY_COIL-<br>Connect K1 pin 16 to RELAY_COIL- using VIOLET; record initials ____.                                                                                  |
| 3    | 90       | VIOLET | K1 pin 1         | P6KE6.8CA lead 1              | Wire              | Bidirectional coil TVS<br>Connect K1 pin 1 to RELAY_COIL+ using VIOLET; record initials _____. Connect D3 lead 1 to RELAY_COIL+ using VIOLET; record initials ____.  |
| 4    | 91       | VIOLET | P6KE6.8CA lead 2 | K1 pin 16                     | Wire              | Bidirectional coil TVS<br>Connect K1 pin 16 to RELAY_COIL- using VIOLET; record initials _____. Connect D3 lead 2 to RELAY_COIL- using VIOLET; record initials ____. |
| 5    | NEW C7-5 | WHITE  | K1 pin 4         | LOCAL_NC                      | Wire              | Connect K1 pin 4 to LOCAL_NC using WHITE; record initials ____.                                                                                                      |
| 6    | NEW C7-6 | WHITE  | K1 pin 6         | LOCAL_TX_SELECTED             | Wire              | Connect K1 pin 6 to LOCAL_TX_SELECTED using WHITE; record initials ____.                                                                                             |
| 7    | NEW C7-7 | WHITE  | K1 pin 8         | LOCAL_NO                      | Wire              | Connect K1 pin 8 to LOCAL_NO using WHITE; record initials ____.                                                                                                      |
| 8    | 105      | BLACK  | K1 pin 11        | GND                           | Wire              | Feedback COM<br>Connect K1 pin 11 to GND using BLACK; record initials ____.                                                                                          |
| 9    | 106      | GREEN  | K1 pin 13        | DevKit GPIO4                  | Wire              | NC feedback<br>Connect K1 pin 13 to K1_NC_FB using GREEN; record initials _____. Connect DEVKIT GPIO4 to K1_NC_FB using GREEN; record initials ____.                 |
| 10   | 109      | GREEN  | K1 pin 9         | DevKit GPIO5                  | Wire              | NO feedback<br>Connect K1 pin 9 to K1_NO_FB using GREEN; record initials _____. Connect DEVKIT GPIO5 to K1_NO_FB using GREEN; record initials ____.                  |
| 11   | 107      | BLUE   | LOGIC_3V3        | R_FB_NC 10 k $\Omega$ , end 1 | Wire              | NC pull-up<br>Connect R_FB_NC 1 to LOGIC_3V3 using BLUE; record initials ____.                                                                                       |
| 12   | 108      | GREEN  | R_FB_NC end 2    | DevKit GPIO4                  | Wire              | NC pull-up node<br>Connect R_FB_NC 2 to K1_NC_FB using GREEN; record initials _____. Connect DEVKIT GPIO4 to K1_NC_FB using GREEN; record initials ____.             |
| 13   | 110      | BLUE   | LOGIC_3V3        | R_FB_NO 10 k $\Omega$ , end 1 | Wire              | NO pull-up<br>Connect R_FB_NO 1 to LOGIC_3V3 using BLUE; record initials ____.                                                                                       |
| 14   | 111      | GREEN  | R_FB_NO end 2    | DevKit GPIO5                  | Wire              | NO pull-up node<br>Connect R_FB_NO 2 to K1_NO_FB using GREEN; record initials _____. Connect DEVKIT GPIO5 to K1_NO_FB using GREEN; record initials ____.             |

### Unpowered test

| Instruction / kind / points / expected                                                                                                             | Evidence |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Map relay before attaching local fixture · continuity · pins 1–16, 4–6, 8–6, 13–11, 9–11 · coil about 50 $\Omega$ ; NC pairs closed; NO pairs open |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                                                                    | Measured |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Exercise bounded relay and fail sources · +5V_RLY/RELAY_COIL- · local contacts/feedback · record current, voltage, VCE, timing, temperature · coil 90–110 mA, voltage at least 4.50 V, BC337 VCE no more than 0.30 V; (1,0) energized, (0,1) bypass; 00 or 11 fault |          |

**Bound operator part record:** K1/D3/JP2/RFB · G5V-2 DC5; P6KE6.8CA; COIL POWER; pull-ups · DPDT 5 V; bidirectional TVS; removable; two 10 k $\Omega$  · 39-G5V-2DC5-ND; 18-P6KE6.8CACT-ND; NOT ORDERED · bottom-view drawing; meter-map physical pins; TVS bidirectional

**Relay exerciser identity:** \_\_\_\_\_ **Firmware identity:** \_\_\_\_\_

From energized state, removal of USB logic power must release to NC bypass; removal of GPIO21 command must release to NC bypass; removal of TREAD\_OK must release to NC bypass; removal of VIN must release to NC bypass. Each NC bypass return must occur in no more than 100 ms. Then hold five minutes: TPS709 and BC337 at or below 45°C and no more than 10°C over ambient.

### Dedicated fail-release and thermal evidence

| Evidence field           | Measured | Limit                                    |
|--------------------------|----------|------------------------------------------|
| USB logic release (ms)   |          | NC bypass $\leq$ 100 ms                  |
| GPIO21 release (ms)      |          | NC bypass $\leq$ 100 ms                  |
| TREAD_OK release (ms)    |          | NC bypass $\leq$ 100 ms                  |
| VIN release (ms)         |          | NC bypass $\leq$ 100 ms                  |
| Ambient temperature (°C) |          | Record before hold                       |
| TPS709 temperature (°C)  |          | $\leq$ 45°C and $\leq$ 10°C over ambient |
| BC337 temperature (°C)   |          | $\leq$ 45°C and $\leq$ 10°C over ambient |

**STOP gate:** Any electrical, feedback, release, reset, or five-minute thermal gate fails. Cluster 7 is local-only; it makes no end-to-end RJ45 claim.

**PASS gate:** Local contacts, feedback, current, voltage, release, and thermal evidence all pass. **PASS — continue** to Cluster 8 only.

Operator: \_\_\_\_\_ Date: \_\_\_\_\_ Signed PASS: \_\_\_\_\_  
Cluster 7 · Local relay proof only



## Cluster 8 — AHC126 and UART taps

**Dependencies:** Cluster 3, Cluster 5, Cluster 7  
**Inputs:** LOGIC\_3V3, TX\_GATE, GPIO17\_TX, LOCAL\_NC, LOCAL\_NO, GND

**Outputs:** TX\_DRV, UART\_BUFFER\_HIGH\_Z, UART\_BUFFER\_FOLLOWS, GPIO18\_RX\_LOCAL, GPIO16\_RX\_LOCAL, LOCAL\_TX\_SELECTED  
**Source state:** Local future-interface fixture only; treadmill and both RJ45 breakouts absent; bounded UART/relay exerciser identity recorded

### Parts

**U6** SN74AHC126N PDIP-14 · 296-4535-5-ND · notch verified. **C6** 100 nF · BC1084CT-ND. **R3** 100 Ω and **R4/R5** 10 kΩ · NOT ORDERED.

**Operator part record:** U6/C6/R3/R4/R5 · SN74AHC126N; capacitor; series/tap resistors · PDIP-14; 100 nF; 100 Ω; two 10 kΩ · 296-4535-5-ND; BC1084CT-ND; NOT ORDERED · notch verified; capacitor non-polar

**Operator wiring record:** See numbered point-to-point steps below.

### Build — exact pin/net/color wiring

| Step | Ref       | Color   | From                | To                     | Part    | Note                                                                                                                                                     |
|------|-----------|---------|---------------------|------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 92        | BLUE    | LOGIC_3V3           | AHC126 pin 14          | Wire    | VCC<br>Connect U6 pin 14 to LOGIC_3V3 using BLUE; record initials ____.                                                                                  |
| 2    | 93        | BLACK   | AHC126 pin 7        | GND                    | Wire    | Ground<br>Connect U6 pin 7 to GND using BLACK; record initials ____.                                                                                     |
| 3    | 94        | YELLOW  | TX_GATE             | AHC126 pin 1           | Wire    | Gate 1 OE<br>Connect U6 pin 1 to TX_GATE using YELLOW; record initials ____.                                                                             |
| 4    | 95        | YELLOW  | DevKit GPIO17       | AHC126 pin 2           | Wire    | Gate 1 input<br>Connect U6 pin 2 to GPIO17_TX using YELLOW; record initials ____.                                                                        |
| 5    | 96        | YELLOW  | AHC126 pin 3        | R_TX 100 Ω, end 1      | Wire    | Gate 1 output<br>Connect U6 pin 3 to TX_BUF using YELLOW; record initials _____. Connect R_TX 1 to TX_BUF using YELLOW; record initials ____.            |
| 6    | NEW C8-7  | YELLOW  | R_TX 2              | TX_DRV                 | Wire    | Connect R_TX 2 to TX_DRV using YELLOW; record initials ____.                                                                                             |
| 7    | 97a       | BLACK   | AHC126 pin 4        | GND                    | Wire    | Unused enables/inputs low legacy Ref 97; 6 wires<br>Connect U6 4 to GND using BLACK; record initials ____.                                               |
| 8    | 97b       | BLACK   | AHC126 pin 5        | GND                    | Wire    | Unused enables/inputs low legacy Ref 97; 6 wires<br>Connect U6 5 to GND using BLACK; record initials ____.                                               |
| 9    | 97c       | BLACK   | AHC126 pin 9        | GND                    | Wire    | Unused enables/inputs low legacy Ref 97; 6 wires<br>Connect U6 9 to GND using BLACK; record initials ____.                                               |
| 10   | 97d       | BLACK   | AHC126 pin 10       | GND                    | Wire    | Unused enables/inputs low legacy Ref 97; 6 wires<br>Connect U6 10 to GND using BLACK; record initials ____.                                              |
| 11   | 97e       | BLACK   | AHC126 pin 12       | GND                    | Wire    | Unused enables/inputs low legacy Ref 97; 6 wires<br>Connect U6 12 to GND using BLACK; record initials ____.                                              |
| 12   | 97f       | BLACK   | AHC126 pin 13       | GND                    | Wire    | Unused enables/inputs low legacy Ref 97; 6 wires<br>Connect U6 13 to GND using BLACK; record initials ____.                                              |
| 13   | 98a       | NO WIRE | AHC126 pin 6        | NO CONNECTION          | NO WIRE | Unused outputs legacy Ref 98; 3 intentional opens<br>Leave U6 pin 6 at NC using NO WIRE; record initials ____.                                           |
| 14   | 98b       | NO WIRE | AHC126 pin 8        | NO CONNECTION          | NO WIRE | Unused outputs legacy Ref 98; 3 intentional opens<br>Leave U6 pin 8 at NC using NO WIRE; record initials ____.                                           |
| 15   | 98c       | NO WIRE | AHC126 pin 11       | NO CONNECTION          | NO WIRE | Unused outputs legacy Ref 98; 3 intentional opens<br>Leave U6 pin 11 at NC using NO WIRE; record initials ____.                                          |
| 16   | 99        | BLUE    | AHC126 pin 14       | C_AHC126 100 nF, end 1 | Wire    | Place close to IC<br>Connect U6 pin 14 to LOGIC_3V3 using BLUE; record initials _____. Connect C_AHC126 1 to LOGIC_3V3 using BLUE; record initials ____. |
| 17   | 100       | BLACK   | C_AHC126 end 2      | AHC126 pin 7           | Wire    | Decoupling return<br>Connect U6 pin 7 to GND using BLACK; record initials _____. Connect C_AHC126 2 to GND using BLACK; record initials ____.            |
| 18   | NEW C8-19 | WHITE   | R_CONS_RX 1         | LOCAL_CONSOLE_6_STUB   | Wire    | Connect R_CONS_RX 1 to LOCAL_CONSOLE_6_STUB using WHITE; record initials ____.                                                                           |
| 19   | NEW C8-20 | GREEN   | R_CONS_RX 2         | GPIO18_RX_LOCAL        | Wire    | Connect R_CONS_RX 2 to GPIO18_RX_LOCAL using GREEN; record initials ____.                                                                                |
| 20   | NEW C8-21 | GREEN   | DEVKIT GPIO18       | GPIO18_RX_LOCAL        | Wire    | Connect DEVKIT GPIO18 to GPIO18_RX_LOCAL using GREEN; record initials ____.                                                                              |
| 21   | NEW C8-22 | WHITE   | R_PIN3_RX 1         | LOCAL_PIN3_STUB        | Wire    | Connect R_PIN3_RX 1 to LOCAL_PIN3_STUB using WHITE; record initials ____.                                                                                |
| 22   | NEW C8-23 | GREEN   | R_PIN3_RX 2         | GPIO16_RX_LOCAL        | Wire    | Connect R_PIN3_RX 2 to GPIO16_RX_LOCAL using GREEN; record initials ____.                                                                                |
| 23   | NEW C8-24 | GREEN   | DEVKIT GPIO16       | GPIO16_RX_LOCAL        | Wire    | Connect DEVKIT GPIO16 to GPIO16_RX_LOCAL using GREEN; record initials ____.                                                                              |
| 24   | NEW C8-25 | WHITE   | K1 pin 4 / LOCAL_NC | LOCAL_CONSOLE_6_STUB   | Wire    | Connect K1 pin 4 / LOCAL_NC to LOCAL_CONSOLE_6_STUB using WHITE; record initials ____.                                                                   |
| 25   | NEW C8-26 | YELLOW  | K1 pin 8 / LOCAL_NO | TX_DRV                 | Wire    | Connect K1 pin 8 / LOCAL_NO to TX_DRV using YELLOW; record initials ____.                                                                                |
| 26   | NEW C8-27 | YELLOW  | K1 pin 6            | LOCAL_TX_SELECTED      | Wire    | Connect K1 pin 6 to LOCAL_TX_SELECTED using YELLOW; record initials ____.                                                                                |

Unpowered test

| Instruction / kind / points / expected                                                                                                                    | Evidence |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Check local buffer map · continuity · U6 pins, 100 Ω resistor, local TX_DRV test point · exact map; disabled output isolated; no connector-side wiring    |          |
| Verify GPIO18 local receive tap · resistance · LOCAL_CONSOLE_6_STUB → GPIO18_RX_LOCAL · 10 kΩ nominal (9–11 kΩ); isolated from LOGIC_3V3, GND, and TX_DRV |          |
| Verify GPIO16 local receive tap · resistance · LOCAL_PIN3_STUB → GPIO16_RX_LOCAL · 10 kΩ nominal (9–11 kΩ); isolated from LOGIC_3V3, GND, and TX_DRV      |          |

Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                                                 | Measured |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Exercise local buffer enable · TX_GATE/GPIO17_TX/local fixture · TX_DRV/UART_BUFFER_HIGH_Z/UART_BUFFER_FOLLOWS · capture logic traces · pin 3 high impedance disabled and follows pin 2 enabled                                                  |          |
| Stimulate GPIO18 local receive tap · LOCAL_CONSOLE_6_STUB low/high from bounded UART exerciser · GPIO18_RX_LOCAL · record low/high voltage and observed bits · low ≤0.40 V; high ≥2.40 V; bits follow                                            |          |
| Stimulate GPIO16 local receive tap · LOCAL_PIN3_STUB low/high from bounded UART exerciser · GPIO16_RX_LOCAL · record low/high voltage and observed bits · low ≤0.40 V; high ≥2.40 V; bits follow                                                 |          |
| Exercise bounded local relay transfer · TX_DRV/K1 LOCAL_NC/LOCAL_NO with bounded relay exerciser · LOCAL_TX_SELECTED · record NC/NO selected resistance and TX waveform · selected path below 2 Ω; unselected path open; waveform follows TX_DRV |          |

Relay exerciser identity: \_\_\_\_\_ Firmware identity: \_\_\_\_\_

Dedicated local future-interface evidence

| Evidence field           | Measured | Limit/state   |
|--------------------------|----------|---------------|
| GPIO18_RX_LOCAL low (V)  |          | ≤0.40 V       |
| GPIO18_RX_LOCAL high (V) |          | ≥2.40 V       |
| GPIO16_RX_LOCAL low (V)  |          | ≤0.40 V       |
| GPIO16_RX_LOCAL high (V) |          | ≥2.40 V       |
| LOCAL_TX_SELECTED NC (Ω) |          | <2 Ω selected |
| LOCAL_TX_SELECTED NO (Ω) |          | <2 Ω selected |

**STOP gate:** TX\_DRV drives while disabled, enabled waveform does not follow, either receive tap fails, local relay selection fails, any connector-side wire is installed, or exerciser identity is absent.

**PASS gate:** Local high impedance, following, receive taps, and relay-selected TX pass; connector-side checks remain deferred. **PASS — continue** to Cluster 9 only.

Operator: \_\_\_\_\_ Date: \_\_\_\_\_ Signed PASS: \_\_\_\_\_  
Cluster 8 · UART isolation and observation

## Cluster 9 — Indicators and VBUS sensing

**Dependencies:** Cluster 3  
**Inputs:** LOGIC\_3V3, GPIO38, USB\_5V, GND

**Outputs:** POWER\_LED, STATUS\_LED, VBUS\_PRESENT\_N  
**Source state:** USB-only test then standalone-only test; never join VBUS to a local rail

### Parts

**Q2** 2N7000 · 4878-2N7000CT-ND; **D4** green LTL-4233 · 160-1130-ND; **D5** red 151051RS11000 · 732-5016-ND; 330  $\Omega$ , 1 k $\Omega$ , and 10 k $\Omega$  resistors · NOT ORDERED; 100 nF · BC1084CT-ND.

**Operator part record:** Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330  $\Omega$ /1 k $\Omega$ /10 k $\Omega$ ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

**Operator wiring record:** See numbered point-to-point steps below.

**Operator bypass wiring record:** C7 · leads 1/2 · LOGIC\_3V3/GND · BLUE/BLACK

### Build — exact pin/net/color wiring

| Step | Ref | Color  | From                   | To                                     | Part | Note                                                                                                                                                                              |
|------|-----|--------|------------------------|----------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 112 | VIOLET | DevKit 5V/VBUS         | 2N7000 gate                            | Wire | Verify physical lead<br>Connect DEVKIT 5V/VBUS to VBUS_SENSE using VIOLET; record initials _____.<br>Connect Q2 gate to VBUS_SENSE using VIOLET; record initials _____.<br>.      |
| 2    | 113 | VIOLET | DevKit 5V/VBUS         | R_VBUS_DISCHARGE 10 k $\Omega$ , end 1 | Wire | Gate discharge<br>Connect DEVKIT 5V/VBUS to VBUS_SENSE using VIOLET; record initials _____.<br>Connect R_VBUS_DISCHARGE 1 to VBUS_SENSE using VIOLET; record initials _____.<br>. |
| 3    | 114 | BLACK  | R_VBUS_DISCHARGE end 2 | GND                                    | Wire | Discharge return<br>Connect R_VBUS_DISCHARGE 2 to GND using BLACK; record initials _____.<br>.                                                                                    |
| 4    | 115 | BLACK  | 2N7000 source          | GND                                    | Wire | Verify physical lead<br>Connect Q2 source to GND using BLACK; record initials _____.<br>.                                                                                         |
| 5    | 116 | GREEN  | 2N7000 drain           | DevKit GPIO7                           | Wire | VBUS_PRESENT_N<br>Connect Q2 drain to VBUS_PRESENT_N using GREEN; record initials _____.<br>Connect DEVKIT GPIO7 to VBUS_PRESENT_N using GREEN; record initials _____.<br>.       |
| 6    | 117 | BLUE   | LOGIC_3V3              | R_VBUS_PULLUP 10 k $\Omega$ , end 1    | Wire | Drain pull-up<br>Connect R_VBUS_PULLUP 1 to LOGIC_3V3 using BLUE; record initials _____.<br>.                                                                                     |
| 7    | 118 | GREEN  | R_VBUS_PULLUP end 2    | DevKit GPIO7                           | Wire | Pull-up node<br>Connect R_VBUS_PULLUP 2 to VBUS_PRESENT_N using GREEN; record initials _____.<br>Connect DEVKIT GPIO7 to VBUS_PRESENT_N using GREEN; record initials _____.<br>.  |
| 8    | 119 | YELLOW | DevKit GPIO38          | R_STATUS 330 $\Omega$ , end 1          | Wire | Status LED<br>Connect DEVKIT GPIO38 to STATUS_LED using YELLOW; record initials _____.<br>Connect R_STATUS 1 to STATUS_LED using YELLOW; record initials _____.<br>.              |
| 9    | 120 | YELLOW | R_STATUS end 2         | Green LED anode                        | Wire | Long LED lead<br>Connect R_STATUS 2 to LED_GREEN_A using YELLOW; record initials _____.<br>Connect LED_GREEN anode to LED_GREEN_A using YELLOW; record initials _____.<br>.       |
| 10   | 121 | BLACK  | Green LED cathode      | GND                                    | Wire | Short/flat-side lead<br>Connect LED_GREEN cathode to GND using BLACK; record initials _____.<br>.                                                                                 |
| 11   | 122 | BLUE   | LOGIC_3V3              | R_POWER_LED 1 k $\Omega$ , end 1       | Wire | Power LED<br>Connect R_POWER_LED 1 to LOGIC_3V3 using BLUE; record initials _____.<br>.                                                                                           |
| 12   | 123 | YELLOW | R_POWER_LED end 2      | Red LED anode                          | Wire | Long LED lead<br>Connect R_POWER_LED 2 to LED_RED_A using BLUE; record initials _____.<br>Connect LED_RED anode to LED_RED_A using BLUE; record initials _____.<br>.              |
| 13   | 124 | BLACK  | Red LED cathode        | GND                                    | Wire | Short/flat-side lead<br>Connect LED_RED cathode to GND using BLACK; record initials _____.<br>.                                                                                   |
| 14   | 125 | BLUE   | LOGIC_3V3              | C_LOGIC 100 nF, end 1                  | Wire | Logic rail bypass<br>Connect C_LOGIC 1 to LOGIC_3V3 using BLUE; record initials _____.<br>.                                                                                       |
| 15   | 126 | BLACK  | C_LOGIC end 2          | GND                                    | Wire | Bypass return C7 lead 1 to LOGIC_3V3 and C7 lead 2 to GND.<br>Connect C_LOGIC 2 to GND using BLACK; record initials _____.<br>.                                                   |

### Unpowered test

| Instruction / kind / points / expected                                                                                                            | Evidence |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Verify transistor leads and LED polarity · polarity · Q2 G/D/S, both LED anodes/cathodes · exact map; VBUS isolated from all local positive rails |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                          | Measured |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Toggle source and status · USB_5V/LOGIC_3V3/GPIO38 · VBUS_PRESENT_N/LEDs · record truth/voltage · VBUS sense active-low; USB present low, USB absent high; red power and green status indications correct |          |

**Bound operator part record:** Q2/D4/D5/resistors/C7 · 2N7000; green/red LEDs; current/sense resistors; capacitor · MOSFET; 330  $\Omega$ /1 k $\Omega$ /10 k $\Omega$ ; 100 nF · 4878-2N7000CT-ND; 160-1130-ND; 732-5016-ND; BC1084CT-ND; NOT ORDERED · Q2 G/D/S mapped; LED long lead anode; flat side cathode

VBUS sense is active-low and isolated: no connection may ever join VBUS to a local rail.

**STOP gate:** VBUS joins a local rail, active-low truth is wrong, LED polarity/current limiting is wrong, or sources overlap.

**PASS gate:** Both LEDs and isolated active-low VBUS sense pass under exclusive sources. **PASS — continue** to Cluster 10 only.

Operator:

Date:

Signed PASS:

## Cluster 10 — Whole-device standalone bench test

**Dependencies:** Cluster 1, Cluster 2, Cluster 3, Cluster 4, Cluster 5, Cluster 6, Cluster 7, Cluster 8, Cluster 9  
**Inputs:** +8V\_RAW, GND, observer firmware

**Outputs:** standalone boot, Wi-Fi observation, event log, UART idle evidence, verified observer state  
**Source state:** USB physically absent; STANDALONE POWER installed; observer verified; relay off; TX disabled

**Outputs record:** standalone boot · Wi-Fi observation · event log · UART idle evidence

### Parts

**ASSEMBLY** completed Clusters 1–9 · NOT ORDERED · inspect all removable links and polarities. **Firmware artifact** observer image and manifest · NOT ORDERED · exact immutable identity required.

**Operator part record:** ASSEMBLY/FW · completed Clusters 1–9; observer image and manifest · whole device; immutable diagnostic artifact · NOT ORDERED · inspect links/polarities; observer manifest proves relay and TX disabled

**Operator wiring record:** USB · plug · DISCONNECTED · NO WIRE; STANDALONE POWER · between · TSR\_3V3 and LOGIC\_3V3 · BLUE; bench · positive lead · +8V\_RAW · RED; bench · return lead · GND · BLACK

### Build — exact pin/net/color wiring

| Step | Ref       | Color   | From                     | To                    | Part       | Note                                                               |
|------|-----------|---------|--------------------------|-----------------------|------------|--------------------------------------------------------------------|
| 1    | NEW C10-1 | NO WIRE | USB plug                 | DISCONNECTED          | NO WIRE    | Connect USB plug to DISCONNECTED using NO WIRE.                    |
| 2    | NEW C10-2 | BLUE    | STANDALONE POWER between | TSR_3V3 and LOGIC_3V3 | Jumper     | Install STANDALONE POWER between TSR_3V3 and LOGIC_3V3 using BLUE. |
| 3    | NEW C10-3 | RED     | bench positive lead      | +8V_RAW               | Bench lead | Connect the bench positive lead to +8V_RAW using RED.              |
| 4    | NEW C10-4 | BLACK   | bench return lead        | GND                   | Bench lead | Connect the bench return lead to GND using BLACK.                  |

### Unpowered test

| Instruction / kind / points / expected                                                                                                  | Evidence |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------|
| Final standalone source audit · resistance · +8V_RAW/VIN/TSR_3V3/LOGIC_3V3/+5V_RLY to GND · no shorts; USB absent; source handoff exact |          |

### Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                              | Measured |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Boot observer from raw input · +8V_RAW · standalone boot/Wi-Fi/event log/UART idle · record rails, reset log, GPIO16/GPIO18 idle · VIN 7.20–7.90 V; TSR_3V3 and LOGIC_3V3 3.20–3.40 V; relay off, feedback (0,1), TX disabled |          |

**Firmware identity:** \_\_\_\_\_ **Observer identity:** \_\_\_\_\_

**Observer manifest:** \_\_\_\_\_ (path or SHA proving relay and TX outputs disabled)

### Dedicated UART idle evidence

| Evidence field       | Measured | State                     |
|----------------------|----------|---------------------------|
| GPIO16 UART idle (V) |          | Before UART configuration |
| GPIO18 UART idle (V) |          | Before UART configuration |

**STOP gate:** Identity/manifest missing, rail out of range, reset, Wi-Fi/event-log failure, UART idle missing, relay enables, TX enables, or feedback differs from (0,1).

**PASS gate:** Identified observer boots standalone and all passive observations pass. **PASS — continue** to Cluster 11 only.

Operator: \_\_\_\_\_ Date: \_\_\_\_\_ Signed PASS: \_\_\_\_\_

# Cluster 11 — RJ45 pass-through and treadmill bypass

Cluster 11 is the end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass proof.

**Dependencies:** Cluster 7, Cluster 8, Cluster 10  
**Inputs:** CONSOLE.1–8, MOTOR.1–8, verified observer state

**Outputs:** eight mapped conductors, end-to-end CONSOLE.6 ↔ MOTOR.6 NC bypass, bypass-only treadmill evidence  
**Source state:** USB absent; standalone installed; observer verified; relay off; TX disabled; source → state → harness → direct-path sequence

## Parts

**J1/J2** CONSOLE/MOTOR RJ45 breakouts · NOT ORDERED · meter-map terminal numbers. **Harness** reviewed dual-pin fused-DMM current harness · NOT ORDERED. **K1** verified Cluster 7 relay contacts · 39-G5V-2DC5-ND.

**Operator part record:** J1/J2/HARNESS/K1 · CONSOLE/MOTOR RJ45 breakouts; dual-pin fused-DMM harness; verified relay · two 8P8C breakouts; reviewed current harness; G5V-2 · NOT ORDERED; 39-G5V-2DC5-ND · meter-map connector terminals; verify fused jack/polarity; relay bottom-view map proven

**Operator wiring record:** J1/J2 · terminals · unwired · NO WIRE; J1 · pin 1 · GND · BLACK; J2 · pin 1 · GND · BLACK; J1 · pin 7 · GND · BLACK; J2 · pin 7 · GND · BLACK; J1 · pin 2 · +8V\_RAW · RED; J2 · pin 2 · +8V\_RAW · RED; J1 · pin 8 · +8V\_RAW · RED; J2 · pin 8 · +8V\_RAW · RED; CONSOLE.3 · endpoint · MOTOR.3 · WHITE; CONSOLE.4 · endpoint · MOTOR.4 · WHITE; CONSOLE.5 · endpoint · MOTOR.5 · WHITE; CONSOLE.6 · endpoint · K1 pin 4 · YELLOW; MOTOR.6 · endpoint · K1 pin 6 · YELLOW; K1 · pin 8 · TX\_DRV · YELLOW; CONSOLE.6 · through 10 kΩ · GPIO18\_RX · GREEN; PIN3 · through 10 kΩ · GPIO16\_RX · GREEN

## Build — exact pin/net/color wiring

| Step | Ref       | Color   | From            | To      | Part    | Note                                                                                                                                     |
|------|-----------|---------|-----------------|---------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | NEW C11-1 | NO WIRE | J1/J2 terminals | unwired | NO WIRE | Isolated mapping first: keep all J1/J2 terminals unwired; map each conductor independently and verify unrelated pins open using NO WIRE. |

## Independent isolated-map evidence — complete before Step 2

| CONSOLE terminal map |     |        |                    | MOTOR terminal map     |             |                                                                                                                                                  |   | Unrelated pins open / operator initials |
|----------------------|-----|--------|--------------------|------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------------------------------|
| 1                    | 2   | 3      | 4                  | 5                      | 6           | 7                                                                                                                                                | 8 |                                         |
| 1                    | 2   | 3      | 4                  | 5                      | 6           | 7                                                                                                                                                | 8 |                                         |
| Step                 | Ref | Color  | From               | To                     | Part        | Note                                                                                                                                             |   |                                         |
| 2                    | 1   | BLACK  | CONSOLE RJ45 pin 1 | GND                    | Jumper wire | First ground Common only after isolated-map PASS. Connect J1 pin 1 to GND using BLACK; this begins the join pins 1/7 and 2/8 commoning sequence. |   |                                         |
| 3                    | 2   | BLACK  | MOTOR RJ45 pin 1   | GND                    | Jumper wire | First ground Connect J2 pin 1 to GND using BLACK.                                                                                                |   |                                         |
| 4                    | 3   | BLACK  | CONSOLE RJ45 pin 7 | GND                    | Jumper wire | Second ground Connect J1 pin 7 to GND using BLACK.                                                                                               |   |                                         |
| 5                    | 4   | BLACK  | MOTOR RJ45 pin 7   | GND                    | Jumper wire | Second ground Connect J2 pin 7 to GND using BLACK.                                                                                               |   |                                         |
| 6                    | 5   | RED    | CONSOLE RJ45 pin 2 | +8V_RAW                | Jumper wire | First +8 V contact Connect J1 pin 2 to +8V_RAW using RED.                                                                                        |   |                                         |
| 7                    | 6   | RED    | MOTOR RJ45 pin 2   | +8V_RAW                | Jumper wire | First +8 V contact Connect J2 pin 2 to +8V_RAW using RED.                                                                                        |   |                                         |
| 8                    | 7   | RED    | CONSOLE RJ45 pin 8 | +8V_RAW                | Jumper wire | Second +8 V contact Connect J1 pin 8 to +8V_RAW using RED.                                                                                       |   |                                         |
| 9                    | 8   | RED    | MOTOR RJ45 pin 8   | +8V_RAW                | Jumper wire | Second +8 V contact Connect J2 pin 8 to +8V_RAW using RED.                                                                                       |   |                                         |
| 10                   | 9   | WHITE  | CONSOLE RJ45 pin 3 | MOTOR RJ45 pin 3       | Jumper wire | Direct pass-through Connect CONSOLE.3 endpoint to MOTOR.3 using WHITE.                                                                           |   |                                         |
| 11                   | 10  | WHITE  | CONSOLE RJ45 pin 4 | MOTOR RJ45 pin 4       | Jumper wire | Direct pass-through Connect CONSOLE.4 endpoint to MOTOR.4 using WHITE.                                                                           |   |                                         |
| 12                   | 11  | WHITE  | CONSOLE RJ45 pin 5 | MOTOR RJ45 pin 5       | Jumper wire | Direct safety pass-through Connect CONSOLE.5 endpoint to MOTOR.5 using WHITE.                                                                    |   |                                         |
| 13                   | 12  | YELLOW | CONSOLE RJ45 pin 6 | K1 G5V-2 pin 4         | Jumper wire | Relay NC contact Connect CONSOLE.6 endpoint to K1 pin 4 using YELLOW.                                                                            |   |                                         |
| 14                   | 13  | YELLOW | MOTOR RJ45 pin 6   | K1 G5V-2 pin 6         | Jumper wire | Relay COM contact Connect MOTOR.6 endpoint to K1 pin 6 using YELLOW.                                                                             |   |                                         |
| 15                   | 14  | YELLOW | K1 G5V-2 pin 8     | R_TX 100 Ω, end 2      | Jumper wire | Relay NO / ESP transmit Connect K1 pin 8 to TX_DRV using YELLOW.                                                                                 |   |                                         |
| 16                   | 101 | GREEN  | CONSOLE RJ45 pin 6 | R_CONS_RX 10 kΩ, end 1 | Wire        | Console RX tap Connect CONSOLE.6 endpoint to K1 pin 4 using YELLOW. Connect CONSOLE.6 through 10 kΩ to GPIO18_RX using GREEN.                    |   |                                         |
| 17                   | 102 | GREEN  | R_CONS_RX end 2    | DevKit GPIO18          | Wire        | Console RX Connect CONSOLE.6 through 10 kΩ to GPIO18_RX using GREEN.                                                                             |   |                                         |
| 18                   | 103 | GREEN  | RJ45 pin-3 net     | R_PIN3_RX 10 kΩ, end 1 | Wire        | Pin-3 tap Connect PIN3 through 10 kΩ to GPIO16_RX using GREEN.                                                                                   |   |                                         |
| 19                   | 104 | GREEN  | R_PIN3_RX end 2    | DevKit GPIO16          | Wire        | Pin-3 RX Connect PIN3 through 10 kΩ to GPIO16_RX using GREEN.                                                                                    |   |                                         |

## Unpowered test

| Instruction / kind / points / expected                                                                                                                                                                                                                 | Evidence |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Map every connector conductor independently · mapping · CONSOLE.1–8 to MOTOR.1–8 and K1 pins 4/6/8 · each intended path below 2 Ω; post-common cross-shorts absent except pins 1/7 GND and pins 2/8 +8V_RAW; end-to-end CONSOLE.6 ↔ MOTOR.6 through NC |          |

## Powered test

| Instruction / input / output / evidence / limits                                                                                                                                                                                                                                                                                                                                                | Measured |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Run bypass-only treadmill sequence · treadmill +8 V through fused harness · current/drop/receive-only UART/thermal · record source state, harness, restored direct paths, endpoints · separately measure treadmill current no more than 500 mA; supply drop no more than 50 mV; ground-return drop no more than 50 mV; 15-minute endpoints no more than 40°C and no more than 10°C over ambient |          |

**Firmware identity:** \_\_\_\_\_ **Observer identity:** \_\_\_\_\_

**Observer manifest:** \_\_\_\_\_ (path or SHA proving relay and TX outputs disabled)

Bypass-only controlled sequence

- 1. All sources off before RJ45 changes. Attach cables only after Clusters 1–10 signed PASS.
- 2. Verify bypass-only source state: USB absent; STANDALONE POWER installed; observer verified; relay off; TX disabled.
- 3. With all sources off before harness changes, remove both independent direct +8V paths and install the reviewed dual-pin fused-DMM harness. Measure current separately at no more than 500 mA.
- 4. Remove power before removal of the harness. Verify zero current, remove harness, then restore both independent +8V paths and continuity-check each.
- 5. Voltage drop is measured only after direct-path restoration. Thermal test only after direct-path restoration; then run the fifteen-minute hold.

Cluster 11 — dedicated bypass and thermal evidence

| Evidence field          | Measured | Limit/state                        |
|-------------------------|----------|------------------------------------|
| Treadmill current (mA)  |          | ≤500 mA, fused harness             |
| Supply drop (mV)        |          | ≤50 mV after direct paths restored |
| Ground-return drop (mV) |          | ≤50 mV after direct paths restored |

| RJ45 pin          | CONSOLE temperature (°C) | MOTOR temperature (°C) | Limit after 15 minutes |
|-------------------|--------------------------|------------------------|------------------------|
| 1 — GND           |                          |                        | ≤40°C; ≤10°C rise      |
| 2 — +8 V          |                          |                        | ≤40°C; ≤10°C rise      |
| 3                 |                          |                        | ≤40°C; ≤10°C rise      |
| 4                 |                          |                        | ≤40°C; ≤10°C rise      |
| 5                 |                          |                        | ≤40°C; ≤10°C rise      |
| 6 — switched data |                          |                        | ≤40°C; ≤10°C rise      |
| 7 — GND           |                          |                        | ≤40°C; ≤10°C rise      |
| 8 — +8 V          |                          |                        | ≤40°C; ≤10°C rise      |

| Relevant breadboard power endpoint                          | Measured | Limit             |
|-------------------------------------------------------------|----------|-------------------|
| +8V endpoint temperature (°C): hottest exact endpoint _____ |          | ≤40°C; ≤10°C rise |
| GND endpoint temperature (°C): hottest exact endpoint _____ |          | ≤40°C; ≤10°C rise |

The current no-control diagnostic cannot pass the relay exercise gate and cannot pass the UART exercise gate. This is observation only. No treadmill transfer and no MCU TX. A future procedure must require qualified functional-firmware identity and matching production safety evidence; bypass PASS grants no transfer permission.

**STOP gate:** Any map, identity, current, drop, receive-only UART, bypass, thermal, or reset check fails. Power off; do not transfer the relay or transmit.

**PASS gate:** All eight paths, NC bypass, current, restored-path drop, receive-only observation, and thermal evidence pass. **PASS — continue** only to documentation closeout, never to treadmill control.

Operator: \_\_\_\_\_

Date: \_\_\_\_\_

Signed PASS: \_\_\_\_\_